

Weekly Self-Study Tasks to Complete

APIs and GraphQL for Leaders

This document covers all of Week 3 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote API lab, where you will query the API in REST and GraphQL and implement a mutation. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep GraphQL open as you watch so you can run each query the videos describe.

Topic	Material Description	Est. Time	Link / Reference
REST & HTTP	MDN “An overview of HTTP” and “Using Fetch” (official docs) — read and run the examples on methods, status codes, and the fetch request/response model.	~20 min	developer.mozilla.org — HTTP & Fetch
GraphQL Basics	Learning GraphQL (Eve Porcello, LinkedIn Learning, released Jul 17, 2025; 1h 9m course). Watch “What Is GraphQL?” and “GraphQL Queries” (~25 min). Covers the schema/query model and how GraphQL differs from REST. Reinforces the instructor demo.	~15 min	linkedin.com/learning — Learning GraphQL
REST vs GraphQL	How to GraphQL — the free “GraphQL Fundamentals” track (community tutorial). Read and skim how a single GraphQL	~25 min	howtographql.com — Fundamentals

Topic	Material Description	Est. Time	Link / Reference
	endpoint contrasts with multiple REST endpoints. Free.		

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote API — Exploring REST and GraphQL." You will query the same data both ways, then implement a GraphQL mutation. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-api-explorer folder, run npm install, then npm start.
- Open the GraphQL explorer in the browser and the REST endpoint in a terminal.
- Query all quotes over REST and notice the response includes every field (over-fetching).
- Run a GraphQL query that asks for only the fields you need, then add another field.
- Implement the createQuote mutation resolver in src/index.js.
- Run the mutation in GraphQL and confirm the new quote appears in the quotes query.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. The coming weeks put a database behind this API, so focus this week on how clients request and receive data.

Topic	Material Description	Link / Reference
REST	REST APIs (LinkedIn Learning) — a complete course on designing and	linkedin.com/learning — REST APIs

Topic	Material Description	Link / Reference
	consuming RESTful services (Video Course).	
GraphQL	Introduction to GraphQL – the official GraphQL documentation. Schema, queries, mutations, and best practices (Reference / Tutorial).	graphql.org/learn
Fetch	Using the Fetch API – MDN Web Docs. The authoritative guide to requests, responses, and error handling (Reference).	developer.mozilla.org – Using Fetch
Security	OWASP API Security Top 10 – the authoritative list of the most common API security risks (Reference).	owasp.org – API Security Top 10
GraphQL Clients	Apollo Client and TanStack Query docs – how front ends consume GraphQL and REST in production (Reference).	apollographql.com & tanstack.com/query